

Single-Bend Analytic Placement Tests

OPALX element placing sandbox

May 30, 2026

1 Purpose

The sandbox inputs

SBEND placement	sandbox/test-sbend-1/test-sbend-1.in
SBEND vertical focusing	sandbox/test-sbend-2/test-sbend-2.in
RBEND placement	sandbox/test-rbend-1/test-rbend-1.in
RBEND vertical focusing	sandbox/test-rbend-2/test-rbend-2.in

exercise the OPALX-native analytic **SBEND** and **RBEND** without field maps, using the native analytic bend model and explicit poses. The first test of each bend type is a placement consistency test. The second test adds a vertical offset and checks the pole-face/fringe-field focusing against old OPAL 2022.1.

Both tests use explicit element posing:

$$(X, Y, Z, \Theta, \Phi, \Psi),$$

where (X, Y, Z) is the placed body-frame origin and Θ is the body rotation about the global Y -axis in the X - Z drawing plane. The figures use this global X - Z projection; for these horizontal-bend tests the body pose has $Y = 0$, normal to the drawing plane.

2 Bend Models

The two analytic bend elements describe the same physical dipole problem from different geometric conventions. Both are intended to give the design bend angle θ from the integrated dipole field. The distinction is the reference frame used to define the magnet body, the meaning of the length attribute, and therefore the pole-face angles at which the circular reference orbit enters and leaves the magnet.

2.1 Physical distinction

SBEND is a sector bend. The reference orbit is a circular arc and the natural hard-edge pole faces are radial. With no additional pole-face angles the beam enters and exits normal to the pole faces, so no thin-edge focusing is introduced by the geometry itself.

RBEND is a rectangular bend. The magnet body is straight, while the reference orbit is still a circular arc through the dipole field. The orbit therefore crosses the entrance and exit pole faces at nonzero edge angles. For the same θ , this makes edge and fringe focusing part of the natural rectangular-bend model.

2.2 OPALX SBEND model

For the sandbox SBEND tests

$$L_{\text{chord}} = 1.0 \text{ m}, \quad \theta = 45^\circ = \frac{\pi}{4}, \quad R = \frac{L_{\text{chord}}}{2 \sin(\theta/2)} = 1.3065629649 \text{ m}.$$

For parser-created SBEND, OPALX follows old OPAL and treats L as the chord. The placed sector arc length and curvature are therefore

$$L_{\text{ref}} = R\theta = L_{\text{chord}} \frac{\theta}{2 \sin(\theta/2)} = 1.0261721530 \text{ m}, \quad h = \frac{\theta}{L_{\text{ref}}} = \frac{1}{R} = 0.7653668647 \text{ m}^{-1}.$$

OPALX stores this as a **PlanarArcGeometry**. The body-frame origin is at the arc midpoint, so the body coordinate s runs from $-L_{\text{ref}}/2$ to $+L_{\text{ref}}/2$. The local sector orbit is the standard sector-bend reference-orbit convention used in first-order beam dynamics [1, 4]:

$$\mathbf{r}(s) = \begin{pmatrix} \frac{\cos(hs) - 1}{h} \\ 0 \\ \frac{\sin(hs)}{h} \end{pmatrix}, \quad \mathbf{t}(s) = \begin{pmatrix} -\sin(hs) \\ 0 \\ \cos(hs) \end{pmatrix}.$$

For $L_{\text{chord}} = 1 \text{ m}$ and $\theta = 45^\circ$,

$$\mathbf{r}_{\text{entry}} = (-0.0994561837, 0, -0.5000000000) \text{ m}, \quad \mathbf{r}_{\text{exit}} = (-0.0994561837, 0, 0.5000000000) \text{ m}.$$

2.3 OPALX RBEND model

The RBEND tests use the same bend angle and field model as the SBEND tests, but the body geometry is rectangular. The input length is the straight body length,

$$L_{\text{rect}} = 1.0 \text{ m}, \quad \theta = 45^\circ, \quad E_1 = 0, \quad E_2 = \theta.$$

Old OPAL uses the pole-face chord radius for a rectangular bend,

$$R_{\text{CH}} = \frac{L_{\text{rect}}}{\sin E_1 + \sin E_2}, \quad L_{\text{ref}} = |\theta| |R_{\text{CH}}|.$$

For $E_1 = 0$ and $E_2 = \theta = 45^\circ$,

$$R_{\text{CH}} = 1.4142135624 \text{ m}, \quad L_{\text{ref}} = 1.1107207345 \text{ m}, \quad h = 0.7071067812 \text{ m}^{-1}.$$

OPALX stores the magnet body as an **RBendGeometry** with the straight length L_{rect} , while the field chart and reference-particle tracking use the OPAL-compatible pole-face reference path L_{ref} . The body frame is rotated by

$$\Theta_{\text{body}} = -\frac{\theta}{2}.$$

The local coordinate of the entrance edge relative to the rectangular body origin is

$$\mathbf{q}_{\text{entry}} = \begin{pmatrix} -\frac{L_{\text{rect}}}{2} \tan(\theta/4) \\ 0 \\ \frac{L_{\text{rect}}}{2} \end{pmatrix}.$$

With an entrance edge at

$$\mathbf{r}_0 = (X_0, 0, Z_0), \quad X_0 = 0, \quad Z_0 = d,$$

and the OPALX input-file X - Z rotation convention

$$\mathcal{R}_Y(\Theta) \begin{pmatrix} x \\ 0 \\ z \end{pmatrix} = \begin{pmatrix} \cos \Theta x + \sin \Theta z \\ 0 \\ -\sin \Theta x + \cos \Theta z \end{pmatrix},$$

the explicit body pose and exit monitor are

$$\begin{aligned} \mathbf{r}_{\text{body}} &= \mathbf{r}_0 - \mathcal{R}_Y(\Theta_{\text{body}}) \mathbf{q}_{\text{entry}}, & \Theta_{\text{body}} &= -\frac{\theta}{2}, \\ \mathbf{r}_{\text{exit}} &= \mathbf{r}_0 + \begin{pmatrix} -L_{\text{rect}} \tan(\theta/2) \\ 0 \\ L_{\text{rect}} \end{pmatrix}, & \Theta_{\text{exit}} &= -\theta. \end{aligned}$$

For the present 45° geometry,

$$\mathcal{R}_Y(-22.5^\circ) \mathbf{q}_{\text{entry}} = (-0.2832272487, 0, 0.4238795325) \text{ m},$$

so the body pose is shifted upstream of the entrance edge even though the nominal entrance edge itself remains at $z = d$.

2.4 Field and edge model

Both bend types use **ANGLE** as the primary geometric input. **K0** is accepted only as a consistency check against

$$k_0 = \frac{\theta}{L_{\text{norm}}},$$

where L_{norm} is the OPAL-compatible field-normalization length for the specific bend model. The shared OPALX field code evaluates the normalized multipole field in the bend field chart and applies the OPAL-style **FM1DProfile1** Enge fringe envelope. With no explicit **GAP**, the inputs use $g = 2 \text{ HGAP} = 0.04 \text{ m}$, so the default Enge half width is $a = 5g = 0.2 \text{ m}$:

$$\ell_i = \frac{a}{|\cos E_i|}, \quad L_{\text{eff}} = \int_{-\ell_1}^{L_{\text{ref}} + \ell_2} F_{\text{Enge}}(s) ds.$$

The vertical pole-face correction used for the thin-edge estimate is

$$\psi(E) = h_{\text{eff}} \text{HGAP} F_{\text{int}} \frac{1 + \sin^2 E}{\cos E}, \quad k_y(E) = -h_{\text{eff}} \tan(E - \psi(E)).$$

The thin-entry-edge vertical map used to design the second test of each bend family is

$$y_{\text{exit}} = (1 + L_{\text{ref}} k_y(E_1)) y_{\text{entry}} + L_{\text{ref}} y'_{\text{entry}}.$$

2.5 Relation to other codes

code	sector bend convention	rectangular bend convention
OPALX/OPAL	SBEND is tracked as a sector bend, but parser input L is treated as a chord for old-OPAL compatibility and converted to L_{ref} .	RBEND keeps L as the straight rectangular body length and uses the pole-face chord relation R_{CH} for the circular reference path.
MAD-X	SBEND uses the sector-bend convention; L is the reference arc length in the MAD-X sequence model.	RBEND represents a rectangular magnet through edge angles; a rectangular bend can be made sector-like, and a sector bend can be made rectangular-like, by the corresponding choices of E_1 and E_2 [3].
elegant	CSBEND is the high-accuracy canonical sector bend with symplectic integration and detailed edge/fringe options.	RBEN is a rectangular dipole model implemented through sector-bend machinery with edge angles; the manual recommends CSBEND for symplectic tracking [5].

The main practical difference is therefore the length convention. MAD-X and the elegant code generally use an arc-length sector bend as the primary high-order tracking object. OPALX keeps the old OPAL input semantics in the parser: **SBEND** converts the user chord to a sector reference path, while **RBEND** preserves the straight rectangular body length and separately constructs the pole-face reference path used by the field chart.

3 test-sbend-1: placement consistency test

This test checks the first explicitly posed analytic **SBEND** pole-face case: a 45° entrance face, zero exit face, a 0.3 m upstream drift, and the initial particle exactly at $(0,0,0)$. Because the particle is on the reference point, the entrance pole face should not deflect it away from the design exit. The exit monitor shown in Fig. 1 is placed at the exported **B1** exit edge. The input deck derives the body and monitor poses from the same sector geometry used by old OPAL. With

$$\mathbf{r}_0 = (X_0, 0, Z_0), \quad X_0 = 0, \quad Z_0 = d, \quad d = 0.3 \text{ m},$$

the formulas used in `test-sbend-1.in` are

$$\begin{aligned}
 R &= \frac{L_{\text{chord}}}{2 \sin(\theta/2)}, \\
 \mathbf{r}_{\text{body}} &= \mathbf{r}_0 + \begin{pmatrix} R(\cos(\theta/2) - 1) \\ 0 \\ R \sin(\theta/2) \end{pmatrix}, \quad \Theta_{\text{body}} = -\frac{\theta}{2}, \\
 \mathbf{r}_{\text{exit}} &= \mathbf{r}_0 + \begin{pmatrix} R(\cos \theta - 1) \\ 0 \\ R \sin \theta \end{pmatrix}, \quad \Theta_{\text{exit}} = -\theta, \\
 \mathbf{t}_{\text{exit}} &= \begin{pmatrix} -\sin \theta \\ 0 \\ \cos \theta \end{pmatrix}.
 \end{aligned}$$

For the present 45° case this gives

$$\begin{aligned}\mathbf{r}_{\text{entry}} &= (0, 0, 0.3000000000) \text{ m}, \\ \mathbf{r}_{\text{body}} &= (-0.0994561837, 0, 0.8000000000) \text{ m}, \\ \mathbf{r}_{\text{exit}} &= (-0.3826834324, 0, 1.2238795325) \text{ m}, \\ \Theta_{\text{body}} &= -22.5^\circ, \\ \Theta_{\text{exit}} &= -45^\circ.\end{aligned}$$

quantity	value
ANGLE	45°
E1, E2	$45^\circ, 0^\circ$
HGAP, FINT	0.02 m, 0.5
ℓ_1, ℓ_2	0.2828427125, 0.2000000000 m
L_{ref}	1.0261721530 m
L_{eff}	1.0261728620 m
h_{eff}	$0.7653663359 \text{ m}^{-1}$
SBEND body pose	$X = -0.0994561837, Y = 0, Z = 0.8000000000, \Theta = -22.5^\circ$
exit monitor pose	$X = -0.3826834324, Z = 1.2238795325, \Theta = -45^\circ$

The practical placement criterion is based on the monitor local coordinates:

$$r_\perp = \sqrt{\bar{x}^2 + \bar{s}^2} < 5 \times 10^{-5} \text{ m}.$$

4 test-sbend-1: OPAL 2022.1 comparison

The old-OPAL comparison uses `test-sbend-1.opal.in` with the same SBEND length, angle, pole-face angles, gap, and fringe integral. The comparison script runs OPALX and OPAL 2022.1 in separate work directories and compares only the `DesignPath.dat` and `.stat` diagnostics; HDF5 and load-balance files are ignored.

In the 30 cm drift comparison used for these figures, the reference-particle B_y peak agrees exactly at the sampled precision. The `.stat` reference-field support starts at the same longitudinal coordinate and ends within 3×10^{-4} m. The remaining path differences are small compared with the bend scale: the maximum design-path differences are 4.4×10^{-4} m in x and 2.1×10^{-4} m in relative z .

5 test-sbend-2: OPALX/OPAL vertical edge-focusing comparison

This test keeps the same chord-to-arc SBEND placement formulas as `test-sbend-1`, but launches a particle with

$$y_0 = 1.0 \text{ mm}, \quad y'_0 = 0.$$

The OPALX input is `sandbox/test-sbend-2/test-sbend-2.in`; the old OPAL reference input is `sandbox/test-sbend-2/test-sbend-2.opal.in`. OPALX uses explicit element placement, while old OPAL uses `ELEMEDGE`. The old OPAL distribution is replicated to 10^4 identical macro-particles so that OPAL 2022.1 writes the standard `.stat` diagnostics; OPALX uses a single particle and no self field.

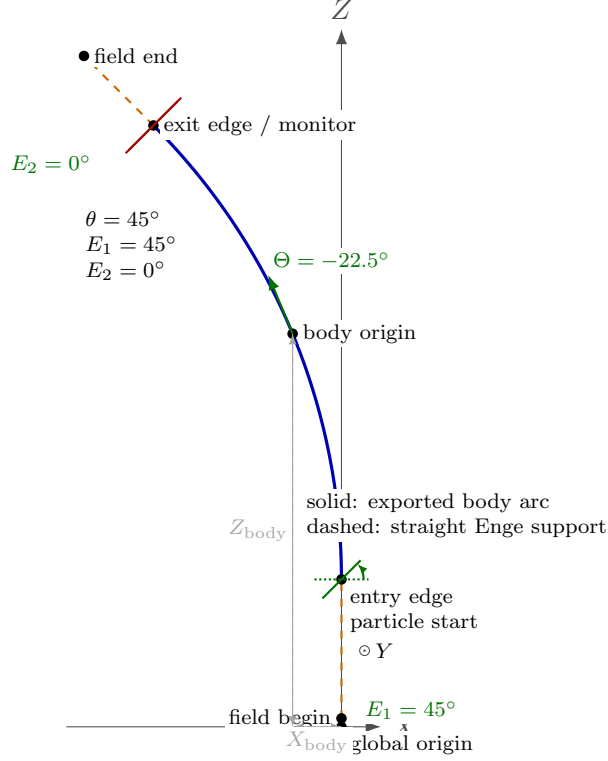


Figure 1: `test-sbend-1`: placed body arc, field support, and temporal monitor plane. The grey dimensions mark the OPALX body pose in the global X - Z projection; the circled-dot axis marks global $+Y$ normal to the page. The entrance pole face is 45° , but the on-axis start particle remains on the design exit.

quantity	value
upstream drift	0.4 m
ANGLE	45°
E1, E2	$-49.99^\circ, 0^\circ$
HGAP, FINT	0.02 m, 0.5
OPAL full GAP	0.04 m
Enge half width $a = 5 \text{ GAP}$	0.2 m
$\ell_{\text{entry}} = a / \cos E_1 $	0.3110798769 m
$\ell_{\text{exit}} = a / \cos E_2$	0.2 m
upstream Enge support start	0.0889199346 m
nominal entrance edge	0.4 m
nominal exit edge	1.4261721530 m

The comparison run uses

$$s_{\text{rel}}^{\text{OPAL}} = s^{\text{OPAL}} - 1 \text{ m}, \quad z_{\text{rel}}^{\text{OPAL}} = z^{\text{OPAL}} - 1 \text{ m}.$$

The apparent bend start near 0.2 m is not a different OPALX element placement. It is the visible part of the upstream Enge fringe. The full OPAL-style support begins at $0.4 - \ell_{\text{entry}} = 0.0889199346 \text{ m}$, where the field is still numerically tiny, and the nominal bend entrance remains at 0.4 m in both OPALX and old OPAL. With the comparison script's 10^{-6} peak-field support threshold, the design-path B_y fringe starts at 0.2401337 m in OPALX and 0.2399838 m in old OPAL.

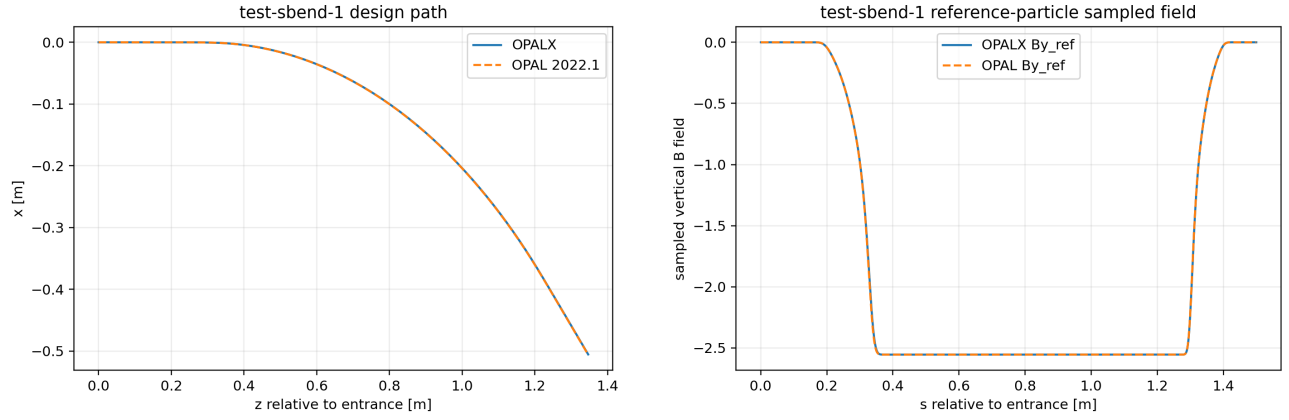


Figure 2: **test-sbend-1** OPALX versus OPAL 2022.1: design-path X-Z orbit (left) and **.stat** reference B_y (right).

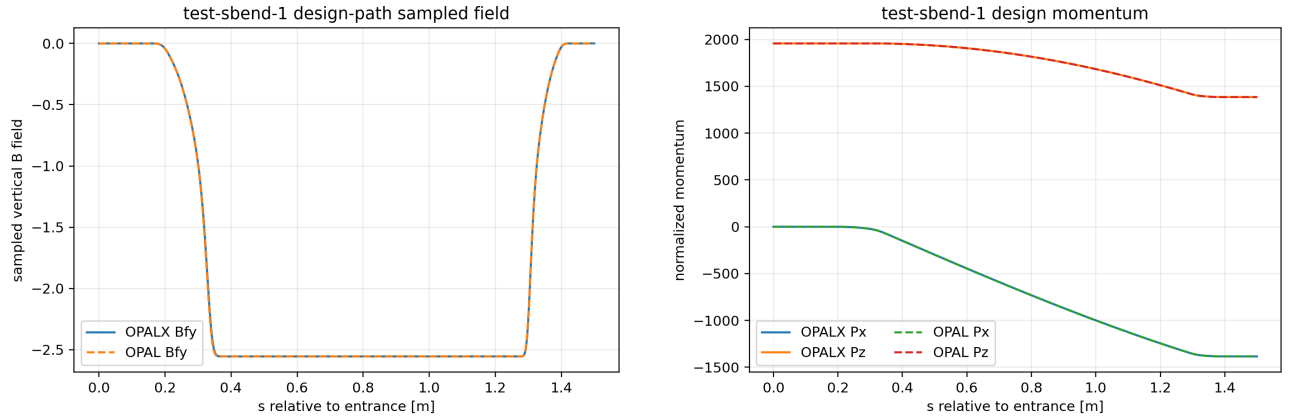


Figure 3: **test-sbend-1** design-path diagnostic comparison. Left: sampled B_y from **DesignPath.dat**. Right: design-path momentum components.

The reference geometry and field profile agree at the same level as the placement test: B_y peak differences are zero in the sampled output, the design-path B_y integral differs by 1.21×10^{-3} , and the `.stat` reference-particle field support ends within 6.0×10^{-4} m. The OPALX vertical edge force is now weighted with the same Enge derivative as the fringe field, so the equivalent kick is applied at the same longitudinal centroid as old OPAL's map. At the final common sample OPALX has $\bar{y} = 2.168074 \times 10^{-3}$ m, while old OPAL has $\bar{y} = 2.161588 \times 10^{-3}$ m. The remaining final difference is 6.49×10^{-6} m, so this case is now a useful vertical-focusing regression diagnostic.

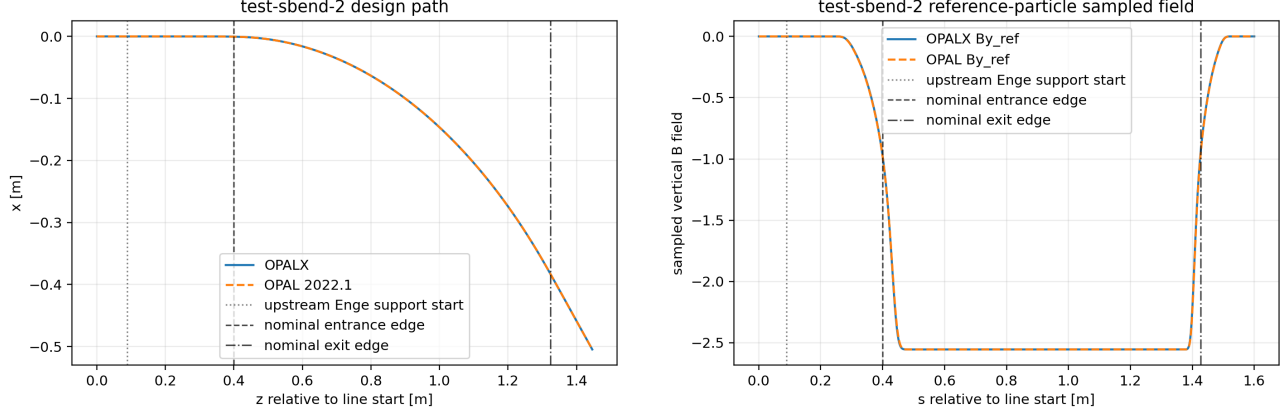


Figure 4: `test-sbend-2` OPALX versus OPAL 2022.1: design-path X-Z orbit (left) and `.stat` reference B_y (right). The dotted vertical marker is the upstream Enge support start; dashed and dash-dotted markers are the nominal entrance and exit edges.

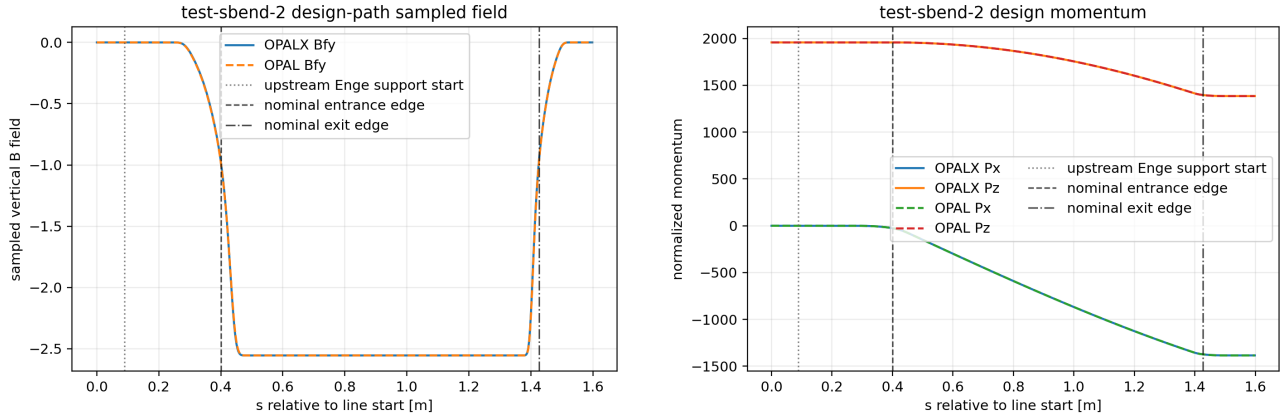


Figure 5: `test-sbend-2` design-path diagnostic comparison. Left: sampled B_y from `DesignPath.dat`. Right: design-path momentum components. The same geometry markers show that the visible fringe precedes the nominal 0.4 m entrance edge in both codes.

6 test-rbend-1: placement consistency test

This test is the RBEND analogue of `test-sbend-1`: a single on-axis particle, a 0.3 m upstream drift, and an explicit RBEND body pose derived from the formulas above. The input decks are

`sandbox/test-rbend-1/test-rbend-1.in`, `sandbox/test-rbend-1/test-rbend-1_opal.in`.

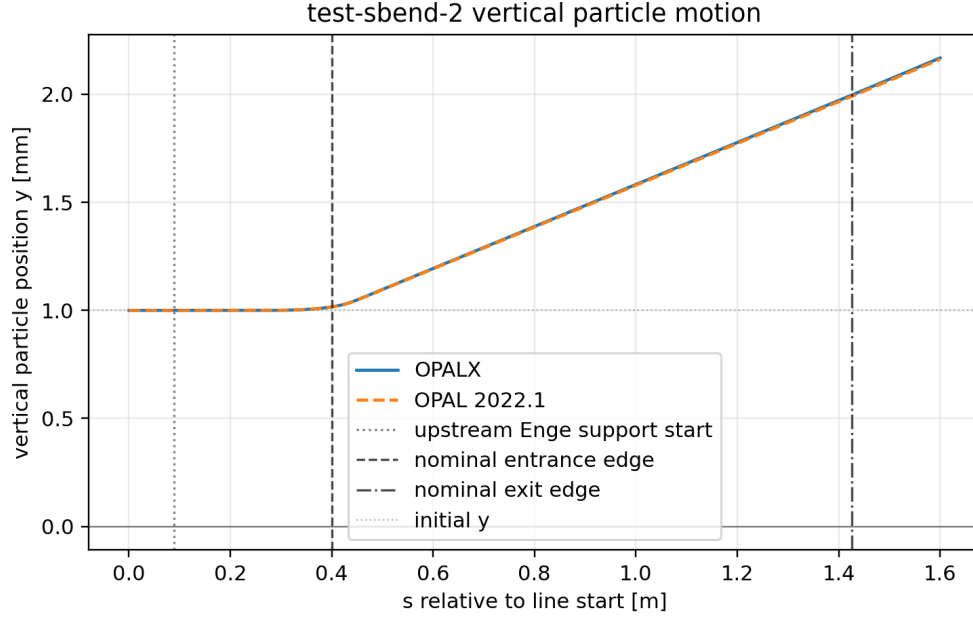


Figure 6: `test-sbend-2` vertical particle motion from the `.stat  $\bar{y}$` column. Both distributions are monoenergetic replicated particles, so $\bar{y}$ is the tracked particle’s vertical coordinate.

The exported OPALX element positions confirm the requested rectangular geometry:

quantity	OPALX export
entrance edge	$X = 5.6 \times 10^{-17}$, $Z = 0.3000000000$ m
body pose	$X = 0.2832272487$, $Z = -0.1238795325$, $\Theta = -22.5^\circ$
exit edge	$X = -0.4142135624$, $Z = 1.3000000000$ m
field support	$Z = 0.1$ to 1.609014865 m

The placement export is therefore consistent with the analytic rectangular survey. With the RBEND field evaluated in the OPAL-compatible rectangular edge frames and normalized with R_{CH} , the physics comparison is now at the same level as the SBEND checks. In the `DesignPath.dat` comparison the largest orbit differences are 2.22×10^{-5} m in X and 2.28×10^{-5} m in relative Z . The sampled B_y peaks are 2.3598578 in OPALX and 2.3598593 in OPAL, and the design-path B_y integral differs by 4.35×10^{-4} . The `.stat` reference-particle field support ends within 3.0×10^{-4} m.

7 test-rbend-2: OPALX/OPAL vertical comparison

This test is the RBEND counterpart of `test-sbend-2`. It uses a 0.4 m upstream drift and launches the particle with

$$y_0 = 1.0 \text{ mm}, \quad y'_0 = 0.$$

The OPALX input is `sandbox/test-rbend-2/test-rbend-2.in`; the old OPAL reference input is `sandbox/test-rbend-2/test-rbend-2_opal.in`. The same rectangular formulas give

$$\begin{aligned} \mathbf{r}_{\text{entry}} &= (0, 0, 0.4000000000) \text{ m}, \\ \mathbf{r}_{\text{body}} &= (0.2832272487, 0, -0.0238795325) \text{ m}, \\ \mathbf{r}_{\text{exit}} &= (-0.4142135624, 0, 1.4000000000) \text{ m}. \end{aligned}$$

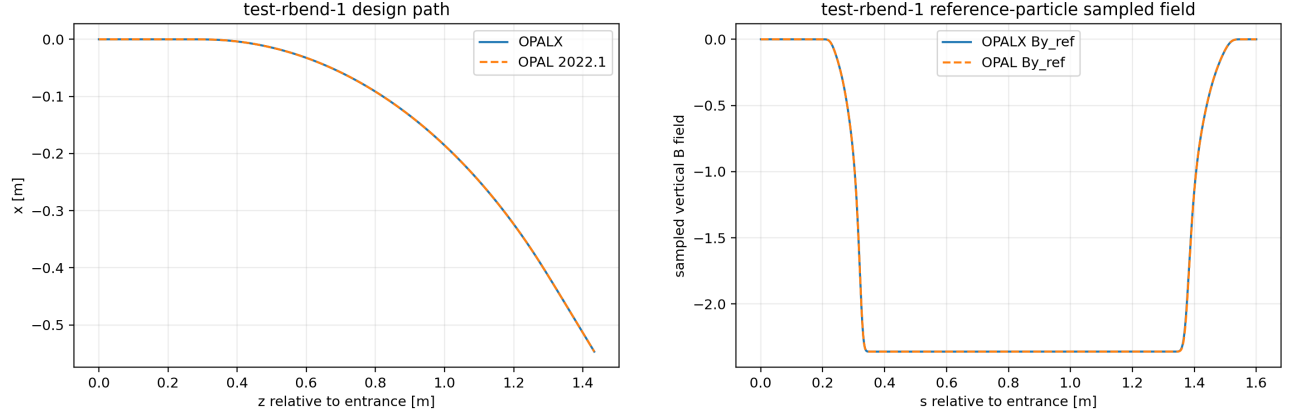


Figure 7: **test-rbend-1** OPALX versus OPAL 2022.1: design-path X-Z orbit (left) and **.stat** reference B_y (right).

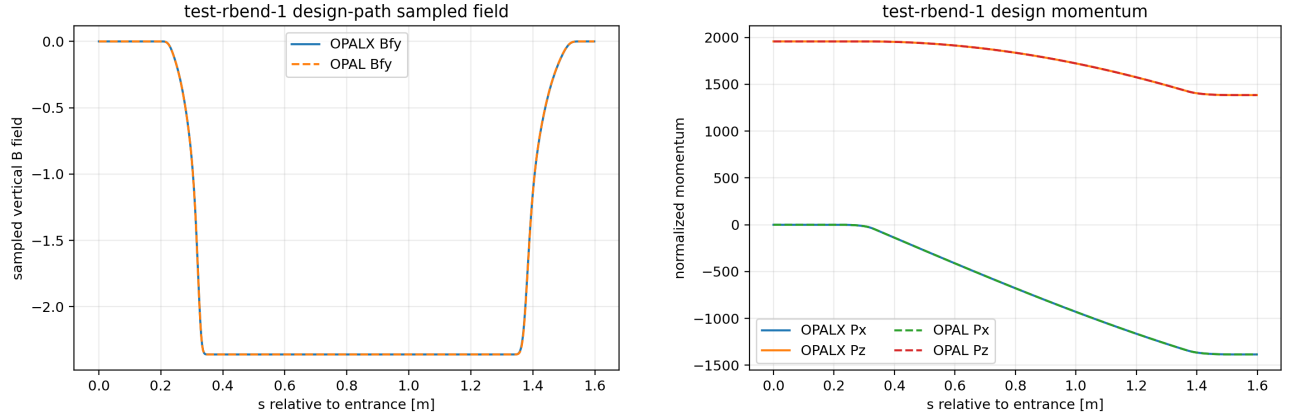


Figure 8: **test-rbend-1** design-path diagnostic comparison. The RBEND B_y peak and longitudinal support now match old OPAL at the sampled precision needed for the placement regression.

The comparison markers are

upstream Enge support start	$s_{\text{rel}} = 0.2000000$	$z_{\text{rel}} = 0.2000000$,
nominal entrance edge	$s_{\text{rel}} = 0.4000000$	$z_{\text{rel}} = 0.4000000$,
nominal exit edge	$s_{\text{rel}} = 1.5107207$	$z_{\text{rel}} = 1.4000000$,
downstream support end	$s_{\text{rel}} = 1.7935634$	$z_{\text{rel}} = 1.6000000$.

The RBEND vertical test now provides a clean edge-focusing regression. The `DesignPath.dat` B_y peaks are 2.3598578 in OPALX and 2.3598593 in OPAL; the design-path B_y integral differs by 4.35×10^{-4} , and the visible field support starts and ends within 1.5×10^{-4} m. At the final common `.stat` sample the mean vertical positions are

$$\bar{y}_{\text{OPALX}} = 8.817085 \times 10^{-4} \text{ m}, \quad \bar{y}_{\text{OPAL}} = 8.804121 \times 10^{-4} \text{ m},$$

for a final difference of 1.30×10^{-6} m. This makes the case suitable as the RBEND counterpart of the `test-sbend-2` vertical edge-focusing regression.

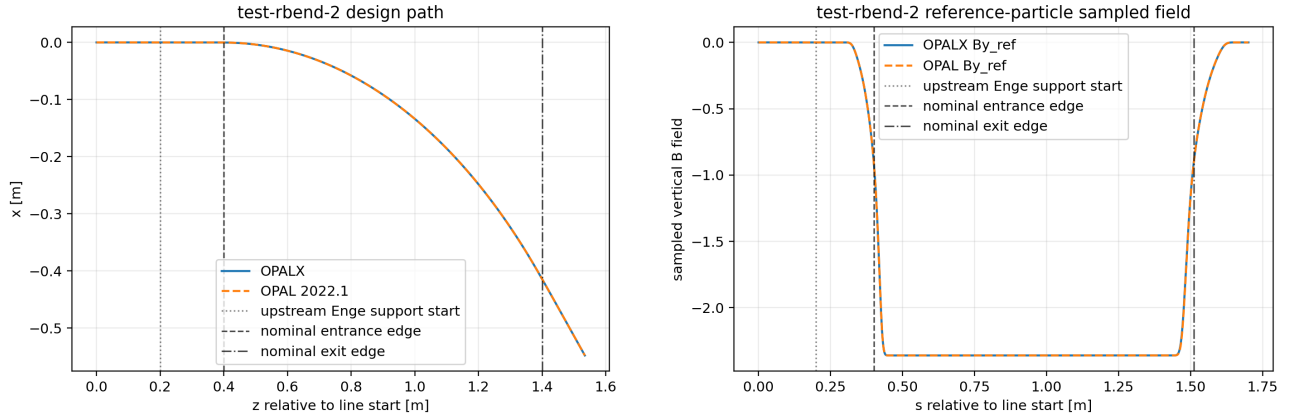


Figure 9: `test-rbend-2` OPALX versus OPAL 2022.1: design-path X-Z orbit (left) and `.stat` reference B_y (right).

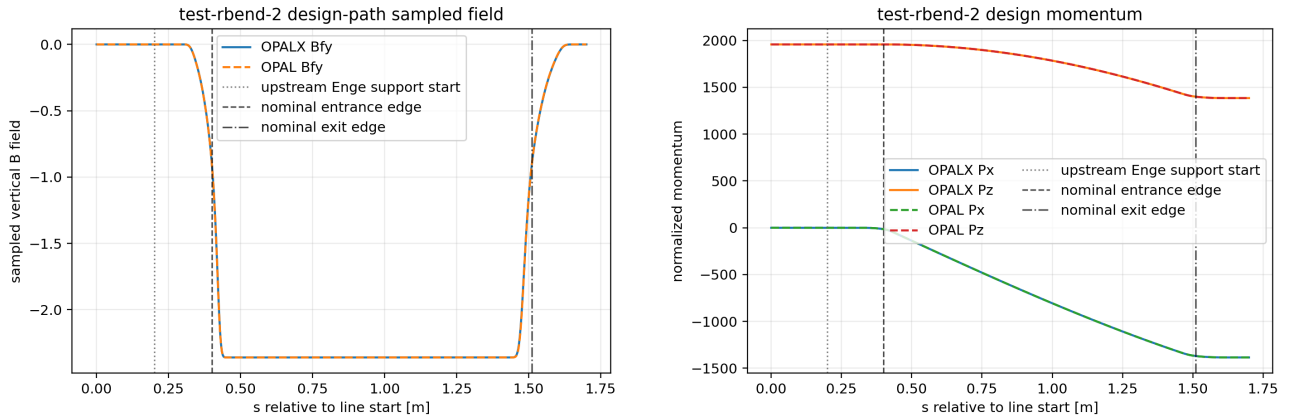


Figure 10: `test-rbend-2` design-path diagnostic comparison. The marker positions use the OPAL pole-face RBEND reference path for s_{rel} and the straight rectangular body length for z_{rel} .

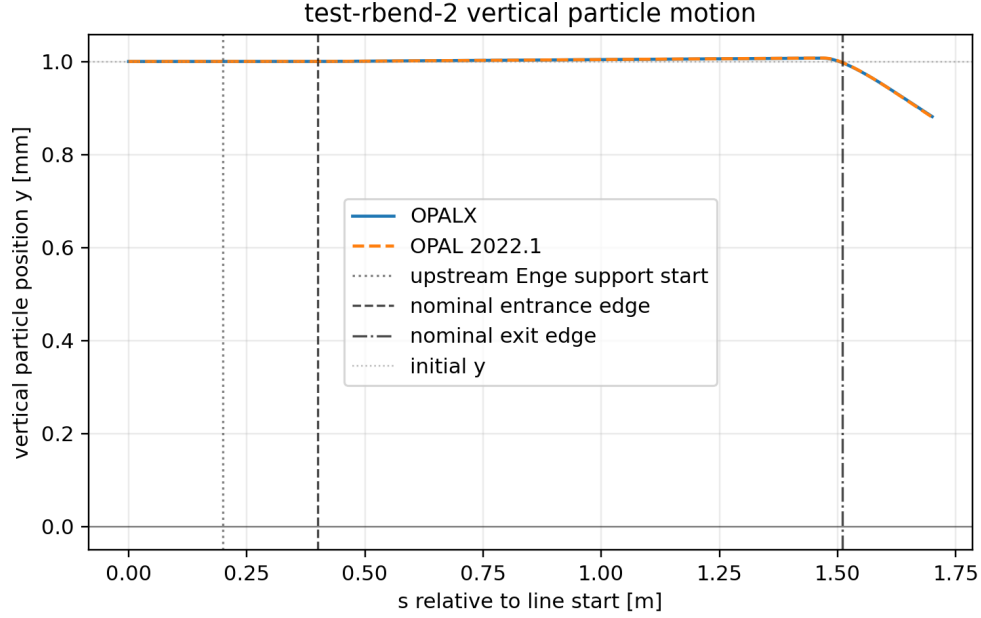


Figure 11: `test-rbend-2` vertical particle motion from the `.stat  $\bar{y}$` column. The final OPALX–OPAL difference is 1.30×10^{-6} m.

8 `test-chicane-1`: symmetric C-shape four-bend chicane

Two sources are useful for a regression design. The USPAS bunch-compression slides give the standard four-rectangular-dipole chicane model and the small-angle expression for R_{56} [6]. The CAS FEL/ERL proceedings discuss the symmetric C-shape chicane as a common FEL compressor geometry and give the same small-angle model together with the usual higher-order longitudinal coefficients [7].

source	useful content	benchmark role
USPAS 2014	four rectangular dipoles, R_{56} , compression map	primary analytic seed
CAS FEL/ERL	C-shape geometry, T_{566} , U_{5666} , timing model	cross-check and context

The OPALX benchmark is a symmetric C-shape chicane with four equal hard-edge dipoles,

$$(+\theta, -\theta, -\theta, +\theta),$$

magnetic length L_b , lateral drift D between dipoles 1–2 and 3–4, and central drift C between dipoles 2–3. To leading order in θ , the central drift does not contribute to the longitudinal dispersion. The school-formula target is therefore

$$R_{56} \simeq -\theta^2 \left(\frac{4}{3} L_b + 2D \right) = -2\theta^2 \left(D + \frac{2}{3} L_b \right).$$

For the same small-angle model one can use

$$T_{566} \simeq -\frac{3}{2} R_{56}, \quad U_{5666} \simeq 2R_{56},$$

as optional higher-order checks once the linear regression is promoted beyond the present linear R_{56} probe.

The executable OPALX realization is

`sandbox/test-chicane-1/test-chicane-1.in.`

It keeps the survey formulas symbolic in the input file. With

$$\rho = \frac{L_b}{2 \sin(\theta/2)},$$

where L_b is the OPAL/OPALX SBEND chord length, not the circular arc length, the reused survey increments are

$$\begin{aligned} \Delta x_b &= \rho(1 - \cos \theta), & \Delta z_b &= \rho \sin \theta, \\ \Delta x_{b/2} &= \rho \left(1 - \cos \frac{\theta}{2}\right), & \Delta z_{b/2} &= \rho \sin \frac{\theta}{2}, \\ \Delta x_{r/2} &= \rho \left(\cos \frac{\theta}{2} - \cos \theta\right), & \Delta z_{r/2} &= \rho \left(\sin \theta - \sin \frac{\theta}{2}\right), \\ \Delta x_D &= D \sin \theta, & \Delta z_D &= D \cos \theta. \end{aligned}$$

The **REAL** symbols **B1_X**, **B1_Z**, ..., **MEXIT_X**, **MEXIT_Z** in the input are assembled from these increments and are then used directly in the element **X**, **Y**, **Z**, and **THETA** attributes.

The old OPAL comparison input is

`sandbox/test-chicane-1/test-chicane-1.opal.in.`

It uses old OPAL 2022.1 **PARALLEL-T** tracking with the same bend signs and the same five-particle momentum scan. In the old OPAL deck the chicane starts at $s = 1$ m; this shifts the absolute monitor **SPOS**, but not the fitted local monitor z slope used for R_{56} .

8.1 Placement result

The OPALX run exports the requested C-chicane survey in the generated element-position text file. The useful placement rows are the **ENTRY EDGE** and **EXIT EDGE** rows for the four bends:

point	z [m]	x [m]
B1 entry	0.000000000	0.000000000
B1 exit / D12 begin	0.998750260	0.049979169
B2 entry	2.491256508	0.199729294
B2 exit / D23 begin	3.490006768	0.249708463
B3 entry	5.490006768	0.249708463
B3 exit / D34 begin	6.488757029	0.199729294
B4 entry	7.981263276	0.049979169
B4 exit / MEXIT	8.980013537	0.000000000

Compared with the symbolic survey formulas above, the largest residuals in the exported OPALX edge table are

$$\max |\Delta z| = 8.28 \times 10^{-10} \text{ m}, \quad \max |\Delta x| = 8.33 \times 10^{-11} \text{ m}.$$

The final exit monitor is therefore centered on the design exit and parallel to the entrance axis at the precision of the exported table.

8.2 R_{56} result

The input carries five particles with

$$\delta = -2 \times 10^{-3}, -10^{-3}, 0, +10^{-3}, +2 \times 10^{-3}.$$

The exit temporal monitor writes one local z coordinate for each particle. The fit uses the local monitor coordinate

$$z_{\text{monitor}}(\delta) - z_{\text{monitor}}(0) \simeq -R_{56} \delta.$$

The local monitor convention has the opposite sign from the usual transport coefficient for this chicane, so the reported number is

$$R_{56} = -\frac{dz_{\text{monitor}}}{d\delta}.$$

case	SPOS [m]	$dz_{\text{monitor}}/d\delta$ [m]	R_{56} [m]	$R_{56} - R_{56}^{\text{target}}$ [m]
small-angle target	—	—	$-4.333333333333 \times 10^{-2}$	—
OPALX	9.001701201	$+4.334628765519 \times 10^{-2}$	$-4.334628765519 \times 10^{-2}$	$-1.295432185213 \times 10^{-5}$
OPAL 2022.1	10.003564372	$+4.337546378253 \times 10^{-2}$	$-4.337546378253 \times 10^{-2}$	$-4.213044919685 \times 10^{-5}$

The OPALX result differs from the old OPAL result by

$$R_{56,\text{OPALX}} - R_{56,\text{OPAL}} = 2.92 \times 10^{-5} \text{ m}.$$

Both codes are close to the small-angle school formula; the remaining difference is expected at this level because the comparison uses finite $\theta = 0.1$ rad, finite integration steps, and a monitor-based local-coordinate extraction rather than a strict first-order transfer map.

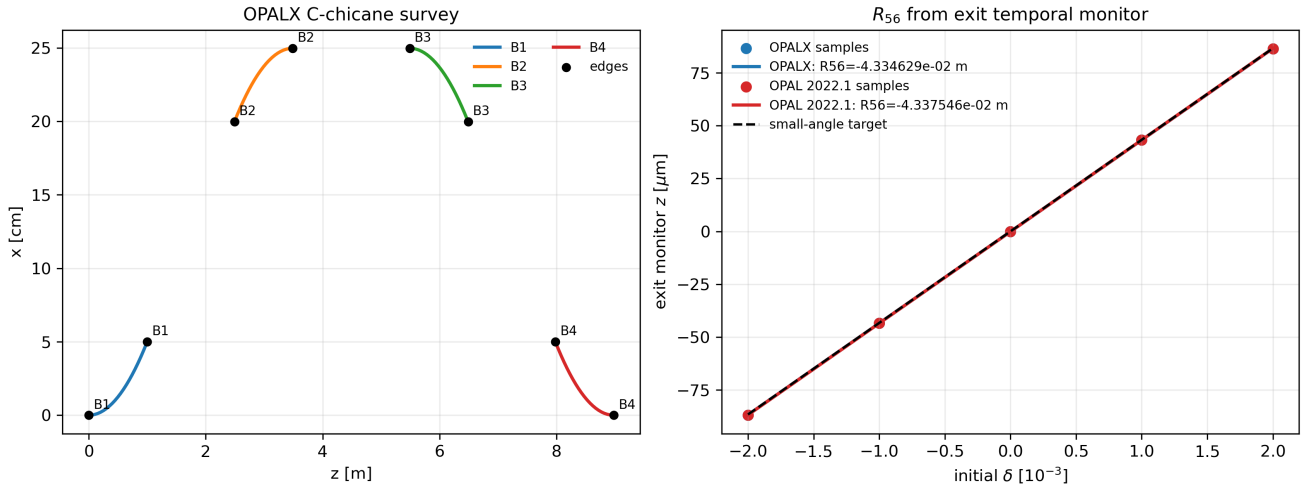


Figure 12: **test-chicane-1** results. Left: OPALX exported C-chicane survey from the bend edge and midpoint rows. Right: exit temporal-monitor z versus initial δ for OPALX and OPAL 2022.1. The dashed line is the small-angle R_{56} target with the monitor-sign convention applied.

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